Assessing household food security in emergencies, does remote data collection cover the gap?

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Abstract

Measuring and reporting global food insecurity is as crucial as addressing the phenomenon itself. However, during emergencies, when reliable and evidence-based food security indicators are needed, it proves difficult to collect data through conventional Face-to-Face (F2F) interviews due to limited field access and security considerations. This has over time created a data gap for food security analysis. With the global surge in mobile and internet access, remote data collection is seen as an important way to bridge the growing data gap in food security analysis. Employing a survey experiment with a stratified sample, this study uses household data from two countries to validate the remote Consolidated Approach for Reporting Indicators of Food Insecurity (rCARI). Comparing the results with the standard Consolidated Approach for Reporting Indicators of Food Insecurity (CARI) methodology reveals two key conclusions. First, rCARI tends to be context-specific. Second, rCARI tends to be conservative relative to CARI by reporting lower food insecurity prevalence. WHile these results show promise in using the remote modules to communicate food insecurity, further empirical studies are needed to fine tune some of the indicators to better align with the standard CARI.

Keywords: Food security, remote, income, expenditure, food consumption score, CARI

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1 Introduction

Food insecurity continue to rise globally posing a significant threat to lives, peace and order. According to the recently publish food security numbers, Measuring food security poses methodological challenges, particularly for organizations operating in populations experiencing food insecurity. Complex data collection and analysis are often impractical, especially in emergency situations (Smith, 2018; Jones et al., 2020). Across various methods, a common practice is the utilization of primary household data gathered through face-to-face interviews or surveys. However, recent events, such as the Covid-19 pandemic and its associated restrictions on movement and physical contact, have significantly hindered conventional data collection approaches. Consequently, the ability to assess the food security needs of populations has been greatly compromised due to the lack of reliable and readily available data (Brown, 2020; White et al., 2021).

Food security assessment and reporting are at the core of WFP's mandate of addressing hunger and food insecurity. Over time, different methodologies (e.g., IPC, CARI, Cadre Harmonise, etc.) have been developed and used to assess and report global food insecurity each differing in scope, and purpose. A commonality that cuts across all methods is the use of primary household data collected through face-to-face interviews/surveys. However, in recent times particularly with the outbreak of the Covid-19 pandemic and attendant restrictions on movement and physical contact, data collection through conventional face-to-face method has been adversely affected. This impacted negatively the ability to assess food security needs because of the data gap created. Developing methodologies to accurately measure and interpret food insecurity especially in emergencies where physical access is restricted thus, is of great importance for WFP and many stakeholders. This is particularly important during emergencies when rapid assessment is required for advocacy to save lives and livelihoods but physical access to the field is restricted either because of health threats or security considerations for enumerators.

To leverage the increased penetration of technology in operational areas of WFP during the Covid lockdowns, a remote version of the Consolidated Approach for Reporting Food Insecurity (CARI)- the so-called rCARI- was developed as an adaptation of the standard CARI methodology and administered remotely (over the phone). The methodology though viewed favourably by researchers, and the larger humanitarian community, because of the stop-gap solution it provided during the covid-19 lockdown, questions about the validity of food security numbers it produces remain. For example, how are the remote modules comparable with already standardized Face-to-Face (F2F) modules in capturing food insecurity. Although there exist validated studies on the use of F2F modules in assessing and reporting on food insecurity, remote methodology such as rCARI remains scant until now.

The potential advantages of the methodology abound for humanitarian work particularly for hard to reach areas. For example, the remote methodology has potential to save lives by providing real-time food insecurity data on the exposed population within the shortest possible time to inform

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policy. Furthermore it can result in better use of scarce resources to achieve high end results by rolling out few targeted modules to assess food insecurity of a larger population.

The overall goal of this study was to assess the validity of the rCARI module(s) against conventional CARI modules for measuring food security within a given population. The study compares indicators collected through phone surveys with conventional face-to-face surveys, considering the latter as the standard methodology. It is important to clarify that the objective of this study is not to report food security at the national level or aim for external validity across the entire population of each pilot country. Instead, the primary goal is to assess the comparability of two methodologies—standard Consolidated Approach for Reporting Food Insecurity (CARI) and remote CARI (rCARI)—when deployed within the same population.

By focusing on the same population within each pilot country, the study seeks to understand how the two methodologies perform under similar conditions and contexts. This approach allows for a direct comparison of the effectiveness and accuracy of the two methodologies in assessing food security indicators.

While the findings of this study provide valuable insights into the relative performance of the two methodologies within specific populations, they may not be representative of food security trends at the national level. Therefore, the conclusions drawn from this study should be interpreted within the context of the population under investigation and may not be generalized to the entire country. By emphasizing internal validity within the selected populations, this study aims to provide robust evidence to inform decision-making and improve food security assessments, particularly in the context of remote data collection during crises such as the COVID-19 pandemic.

The following food security indicators were compared: Food Consumption Score (FCS), reduced Coping Strategy Index (rCSI), Livelihood Coping Strategy (LCS), Food Expenditure Share (FES), and income sources and income change. The Economic Capacity to Meet Essential Needs (ECMEN) indicator, which is recommended to be used to calculate CARI will not be tested in this study, as it requires a standard long expenditure module, which cannot be collected over phone. In addition, ECMEN calculation requires having an updated value of the minimum expenditure basket (MEB) of the study population, which is not available for some of the study areas. Instead, a shortened version of the standard expenditure module will be used to test the possibility of collecting FES data over the phone.

Studying the validity of using the phone to collect food security indicators is a key step in judging the appropriateness of the rCARI methodology to report food insecurity prevalence. However, testing the comparability of these indicators through different modalities of data collection (F2F vs. Phone) should be read within the wider consideration around phone survey modality of data collection. For example, other factors aside from the modules should be considered while reading the results of any remote survey.

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1. Mobile phone ownership (mobile phone penetration rate in the study area) could likely exclude the most vulnerable population from the sample.

2. Mobile phone network coverage; Even in countries with high penetration rates, network coverage could be a problem, especially when electricity cuts are frequent and local infrastructure, such as communication masts, are not evenly distributed (World Bank, 2020).

3. Response rate: It is well documented that the response rate in phone surveys is usually lower than in face-to-face surveys (L'Engle et al., 2018; World Bank, 2020).

Nevertheless, there are several statistical weighting techniques that can adjust the results of phone surveys based on baseline variables like demographics characteristics of the most recent F2F surveys (Nagpal et al. 2021; Glazerman et al. 2020; Valliant et al. 2013; Elkasabi, 2015). This allows the final results of such exercises to be aggregated and extrapolated to reflect some level of representativeness. However, this current study may not be affected by these confounding issues since the main idea is not to report the results at the national level but rather to evaluate the comparability of remote household interviews in collecting and reporting food security. For this reason, the current study report covering the 3 pilot countries does not use any weights. Nonetheless, the study design adopted in Central African Republic (CAR), could allow for testing of sample skewness due to mobile phone ownership and coverage (see CAR sample design in Annex). CAR has one of the lowest mobile phone penetration rates globally (approximately 70% of the population do not own mobile phones). In the next section, the study design and the detailed sampling procedure adopted in each country as well as the study implementation is outlined.

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2 Study design

The study was implemented as a survey experiment with stratified sample of respondents who are randomly selected to two groups; CARI Group (F2F) and rCARI Group (Remote). The study designed followed rigorous randomization procedures to assign potential respondents to the study groups to ensure statistical comparability of the two sub-samples. For a systematic household/respondent selection process, refer to Figure 1.

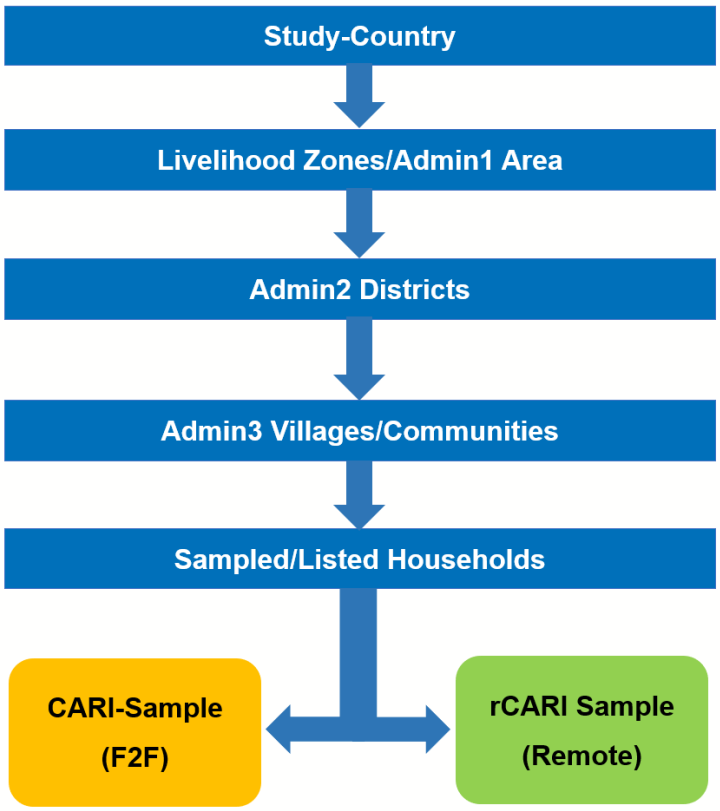

Figure 1: rCARI Validation Study Sampling Frame

1. STEP 1 - Livelihood Zones/ADMIN1 Area Selection: A minimum of 2/3 Admin1 Areas were identified together with WFP CO based on phone coverage, food security situation and accessibility. If more than 3 Admin1 areas met the criteria, a simple random approach was used to select the required number of Admin1 Areas.

2. STEP 2 - ADMIN2 Selection: A minimum of 4 ADMIN2 areas were randomly selected (2 per ADMIN1 Area based on the outcome of Step 1).

3. STEP 3 - ADMIN5 (villages, communities or camps) Areas: All ADMIN 5 (villages) in ADMIN2 that meets the inclusion criteria set out in STEP 1 are now compiled at this stage as the primary sampling units (PSUs). A total of 20 PSUs were randomly selected (distributed

uniformly across the selected ADMIN1 AREAS) as sampling clusters to draw the households from.

4. STEP 4 - Sampled households: The sampling frame for the study was all households in selected PSUs. A minimum of 100 households per PSU were selected through the random-walk approach (listing every 3rd house in the sequence) by trained enumerators. Consent of participation was taken from all HHs before being included into the study sample. Based on the number of PSUs per country, 2,000-2,500 HHs (except in Myanmar where the initial sample was 1,665HHs). The final listed households in this stage were used for the randomization process in the next step.

5. STEP 5 - Randomization of household (hh): Consented final sampled households in STEP 4 are randomly assigned into one of two survey modalities/groups; F2F (CARI group) and Remote (rCARI group). The designed is programmed such that sample will be divided equally between the modalities. All interviews were conducted over the same time frame to ensure that households in the two modalities are reporting their food security situation over the same window.

After the randomization was done (STEP 5), using baseline household and respondents collected during the listing phase, a statistical balance of the two groups is done. This is to ensure that the two study groups are statistically comparable. Food security data was collected using standardized face-to-face and remote modules administered to the two groups collecting household level food security indicators.

The study is being implemented across six countries: 2 in Africa, 2 in Latin America, 1 in Southeast Asia, and 1 in the Middle East. After the first round of data collection covering the period May 2023 to August 2023, surveys in Myanmar, Ecuador and CAR involving approximately 3,755 households have been completed. The remaining 3 countries are planned to start soon.

2.1 Selection of study areas and participants

Survey participants were enlisted into the study through a stratified random process. A minimum of 100 potential participating households were enlisted per Admin5 area (except for Myanmar where the study was conducted amongst 20 IDP camps and some of the camps had fewer than 100 households). The listed households formed the sampling frame for the group/modality assignment which was then done randomly.

In the listing phase of the study, baseline variables that included demographic characteristics of potential respondents and their households was collected in Ecuador and CAR. This was not possible in Myanmar because field access restrictions. The initial sample size listed for the study was 1665 households in Myanmar, 2372 households in Ecuador, and 2500 households in CAR.

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2.2 Myanmar

The study in Myanmar was implemented in two regions: Kachin and Shan (Admin 1). The sampling list was compiled using WFP beneficiaries made up of internally displaced persons (IDPs) in the two regions. Phone ownership was a key criterion for inclusion in the study. However, being WFP beneficiaries, all participants in Myanmar owned phones. A total of 20 camps were selected as the primary sampling units (PSU), respondents/households were then randomly selected and used in the randomization stage.

Data collection was conducted by WFP field staff after a 3-day training that was led by the WFP Regional Bureau of Bangkok in collaboration with the Myanmar CO and supported by the rCARI study team from HQ. The training and data collection were administered in Burmese, the official language of Myanmar.

A total of 610 (73%) household interviews were completed for the F2F modality, while 483 (58%) households were interviewed by phone (remote modality). A balance test of the randomization indicated that the baseline household characteristics were well balanced between the two groups (modalities).

2.3 Ecuador

In Ecuador, the study was implemented in three regions: Amazonia, Coasta and Sierra (Admin1 areas). In addition to the standard selection criteria based on accessibility and food security prevalence, consideration was also given to the safety of the team, considering the political upheaval at the time of implementing the study. A total of 27 Admin5 areas (villages) were randomly selected from these 3 Admin1 areas.

The sampling frame was all households in the 23 ADMIN5 areas that had been randomly selected to participate in the study. The study sample was enlisted following a random walk in each of the Admin5 areas and selecting every 3rd household. Demographic characteristics as well as contact details including the phone number of the household head was then collected at this stage (listing phase). Contact details for a total of 2,374 households were collected after receiving written consent. Since households may be interviewed by phone, potential respondents or someone within the household had to own a phone through which they may be contacted for the survey. Households that were selected but had no phone, were eliminated, and replaced by the next household on the list. This was, however, observed to be minimal in Ecuador because of the high national mobile phone penetration 80% (see Figure 2 for the sample distribution and levels of randomization).

Enumerator training was jointly conducted by the Ecuador CO and the Regional Bureau of Panama (RBP) colleagues in Spanish over a 4-day period with a one-day simulation for remote and F2F instruments. Interviews were conducted in Spanish, the official language of Ecuador. The field work in Ecuador took place between June and July 2023.

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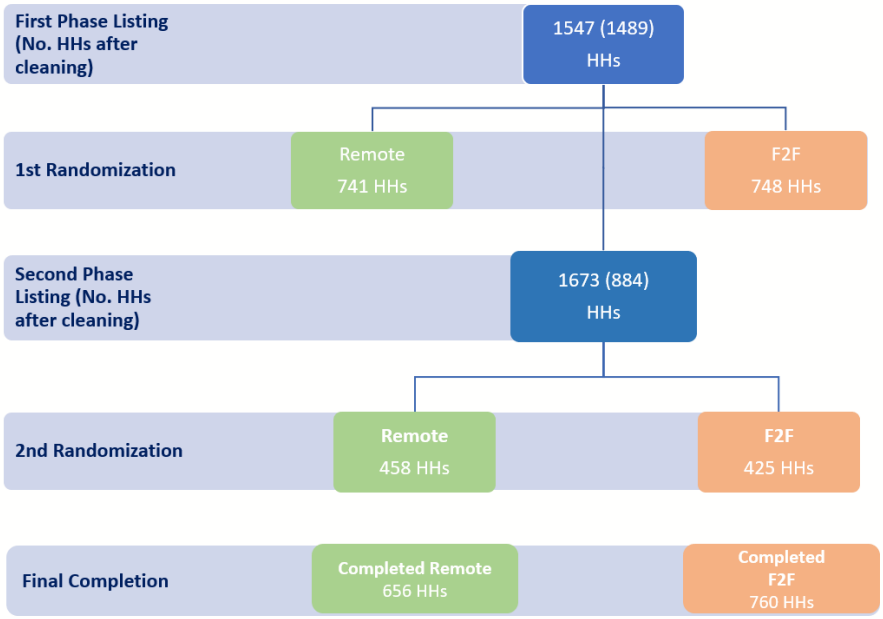

Figure 2: Sample Composition and Completion in Ecuador

2.4 Central African Republic

The study in CAR was implemented in 2 prefectures - Mbomou and Nana-Gribizi (Admin1), the two most accessible areas (aside from Bangui) that also had a varying level of food security. The selection of PSUs followed a similar pattern using field access (security situation) and mobile phone network coverage. The list of Admin5 areas that met the access and mobile phone coverage criteria formed the sampling frame. A total of 25 Admin5 areas (villages) were randomly selected from the list of Admin5 areas compiled jointly by the CAR CO and the local mobile phone service providers (to ensure that all Admin5 areas selected had mobile phone network). Admin5 areas without phone coverage were automatically excluded. To ensure that the final sample was a mix of rural and urban households (taking into consideration there was a potentially different food security situation between rural and urban households), sub layer of locality was included during the selection of Admin5 areas so that the final sampled villages was made up of rural and urban areas. Implementation of the household listing and survey was led by ICASES, the National Statistical Office of CAR.

The sampling frame for the study was all households in the 25 selected ADMIN5 areas. The listing phase was conducted through the random-walk methodology in each selected PSU by trained enumerators.

A minimum of 100 households/respondents per PSU totaling 2500 households/respondents were listed as potential study participants. Phone numbers of the main respondent and any other phone numbers within the household that could have been used to contact the main respondent during

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the survey was collected. Drawing from the data collection experience in Myanmar and Ecuador, a number of demographic characteristics such as age of household head, gender of household head, household size, and ownership of common household assets was collected during household listing and used for the statistical balance test.

Training of enumerators was conducted facilitators from the Regional Bureau of Dakar, and HQ with the support from the CAR CO. The training was conducted in French over a in 4-day period, including one day used for field testing, adjusting and adapting the final questionnaires (F2F & Remote) before deployment for data collection. While the enumerator training was conducted largely in French (except of the simulation), deployment for the actual field work was done systematically by the CO and ICASES so that each enumerator was sent to localities where they could speak the local dialects in case a respondent does not speak French thus require the interview is conducted in the local Sango dialect.

A large proportion of the population of CAR (approximately 70%) do not own mobile phones. To ensure that the study compared appropriate households in terms of phone ownership and coverage, a slightly different sampling and randomization methods were applied (see Figure 2 for all the sampling and randomization stages adopted in CAR). This allowed a segregation of phone-owners and non-phone-owners so that the results are not only compared by modality but also by phone-ownership. A statistical balance test between the different groups showed that respondents were significantly balanced (see Annex for details).

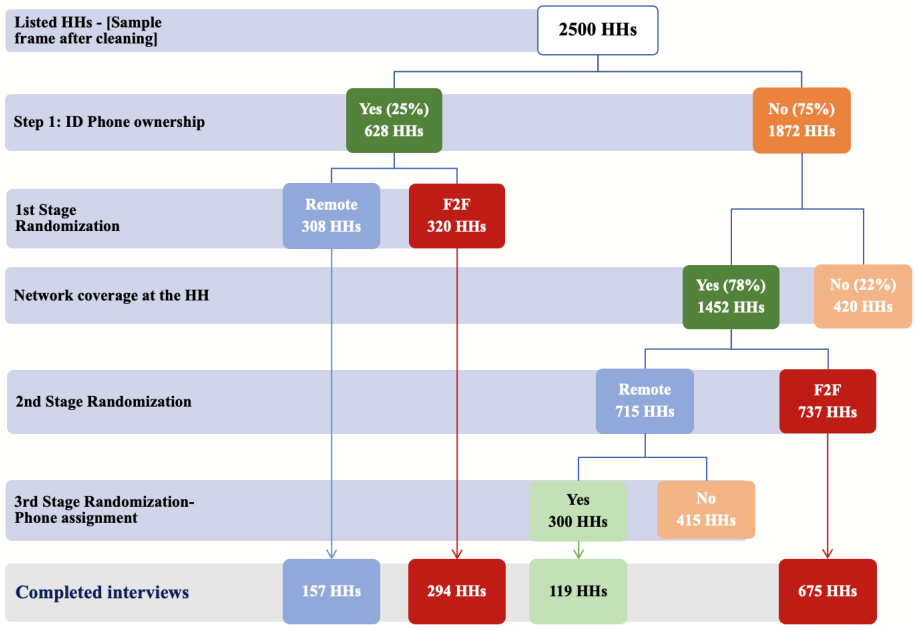

Figure 3: Sample Composition and Completion in CAR

2.5 Overall study implementation

Data collection for both modalities was done simultaneously to cover the same period. The fieldwork was originally planned over a 3-week period (covering 1 week for enumerator training, adjusting questionnaire and 2 weeks for actual household interviews) in each pilot country. However, due to unforeseen challenges, data collection was extended by between 2 to 3 weeks in each country. See Figure 4 on the study implementation timelines and status in each of the pilot countries for the year ending 2023.

Countries / Time	Jan - Apr	May	Jun	Jul	Aug	Sep - Nov	Status
Overall Design & Planning	✓						
Myanmar		Planning	Data Collection				Completed
Ecuador			Planning	Data Collection			Completed
CAR				Planning	Data Collection		Completed
Burkina Faso					Planning Ready	Listing Ready	On-hold
Lebanon							TBC
Mauritania							TBC
Guatemala							TBC
Data Cleaning & Analysis		Pilot	Myanmar	Ecuador	CAR	**Compiling**	Ongoing
Report Writing							Ongoing

Figure 4: rCARI Validation Study Timelines & Potential Pilot Countries

In the next section, sample completion and initial results are presented for all countries completed at this stage of the study.

2.6 Survey Instrument(s)

In this section, a brief description of the survey instruments is provided detailing the differences between F2F and remote surveys. Taking advantage of the unique experimental design of the study, we tested different versions of the standardized modules defined according to the length of the questionnaire, short and long modules. Expenditure and food consumption score were the two modules that had variations for this purpose.

2.6.1 F2F Questionnaire

The Face-to-Face (F2F) questionnaire used in this assessment comprises standard food security assessment modules that have been consistently employed by the World Food Programme (WFP). These modules cover various aspects of household food security, providing comprehensive insights into household demographics, food consumption patterns, coping strategies, livelihoods, expenditures, essential needs, income sources, assets, and the Food Insecurity Experience Scale (FIES)

modules.

The F2F questionnaire is available in two versions: short and long. The version deployed depends on the length of items related to expenditure and food consumption modules. The long questionnaire includes all standard expenditure and food consumption items typically queried in food security assessments. It encompasses a total of 154 questions, covering detailed inquiries into both food and non-food items related to household expenditures.

In contrast, the short version of the questionnaire is a condensed version, aimed at reducing respondent burden and interview duration. It consolidates some expenditure items into groups and aggregates expenditure data for these groups rather than listing individual items. As a result, the short version comprises fewer questions, totaling 108 in all.

2.6.2 Remote Questionnaire

The remote questionnaire is an adapted version of the standard Computer-Assisted Telephone Interviewing (CATI) modules, which have undergone rigorous testing and validation for remote data collection. It includes the same modules as the Face-to-Face (F2F) questionnaire, with some exceptions such as the exclusion of expenditure, asset, and health modules. Typically it is difficult to collect expenditure data in phone interviews due to time constraints and potential respondent fatigue. However, in this experiment, a variant of the standard F2F expenditure module was piloted remotely. This involved condensing and aggregating the number of expenditure items under food and non-food categories, aiming to explore the feasibility of collecting expenditure data remotely.

Another notable difference between the remote and F2F questionnaires lies in the Food Consumption (FC) module. While the remote questionnaire only asks about main food groups, the F2F version includes a comprehensive list of different food items and sub-groups. This difference allows for a more detailed assessment of food consumption patterns in the F2F interviews compared to the remote approach. Overall, the remote questionnaire maintain consistency with the F2F questionnaire while addressing practical considerations and limitations associated with remote data collection.

Similar to the F2F questionnaire, the remote questionnaire also offers two variants: the long and short versions, differentiated by the number of expenditure questions included. The long version of the remote questionnaire consists of 23 questions related to both food and non-food expenditure items. On the other hand, the short version of the remote questionnaire comprises only 5 questions for both food and non-food expenditure items. The idea is to test which version of the two may be close to the standard expenditure module and allow for collecting expenditure data remote for food security assessments.

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2.7 Survey protocols

Before, during, and after the survey, several protocols were established to ensure the successful collection of high-quality data. These protocols were designed to address the challenges of low response rate and to promote transparency, trust, and participation among respondents.

Firstly, collaboration with local authorities and influential individuals in each community (ADMIN5s) was initiated, facilitated by WFP Country Offices (COs) and survey contractors. This engagement aimed to sensitize potential respondents to the survey, fostering transparency and increasing trust. By involving key stakeholders at the local level, the survey team was able to garner support and cooperation from the community, laying the foundation for successful data collection.

Secondly, a stand-alone phase household listing (sampling) was implemented, allowing enumerators to inform households of the upcoming interviews during the listing phase. This proactive approach ensured that households were aware of the survey and had the opportunity to choose a convenient time for the interviews. By providing advance notice and allowing households to schedule interviews at their convenience, the risk of ambushing respondents or calling them at inconvenient times was minimized, enhancing the likelihood of participation and cooperation. Overall, these pre-survey protocols were essential for establishing rapport with the community, building trust among respondents, and ensuring that data collection proceeded smoothly and effectively. By proactively addressing potential challenges and engaging stakeholders at the local level, the survey team was able to maximize response rates and achieve high-quality data outcomes.

In addition to the pre-survey protocols aimed at community engagement and scheduling, a comprehensive calling protocol was developed in collaboration with the hunger monitoring team of the World Food Programme (WFP). Leveraging their expertise in remote data collection, this protocol was designed to ensure consistency, accuracy, and professionalism during phone interviews. The calling protocol outlined guidelines and procedures for enumerators to follow when conducting interviews remotely. These guidelines covered various aspects of the interview process, including:

1. **Introduction:** Enumerators were instructed on how to introduce themselves and the purpose of the survey to respondents in a clear and professional manner, fostering trust and cooperation from the outset.

2. **Consent:** Procedures for obtaining informed consent from respondents were outlined, emphasizing the importance of respecting respondent privacy and autonomy throughout the interview process.

3. **Questioning Techniques:** Enumerators were trained on effective questioning techniques to elicit accurate and comprehensive responses from respondents while maintaining a respectful and non-biased approach.

4. **Data Recording:** Guidelines for accurately recording respondent answers and maintain-

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ing data integrity were provided to ensure consistency and reliability in the collected data.

5. **Ethical Considerations:** Enumerators were reminded of the ethical considerations involved in conducting phone interviews, such as confidentiality and data security, and instructed to adhere to ethical standards throughout the survey process.

Training sessions for enumerators included a detailed review of the calling protocol, with practical exercises and simulations to reinforce understanding and proficiency. By equipping enumerators with clear guidelines and best practices for conducting phone interviews, the calling protocol aimed to enhance the quality and reliability of the collected data, ultimately contributing to the success of the survey.

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3 Data & Empirical Strategy

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4 Results and Discussions

In this section, we present the main results of the study by comparing food security indicators using different versions of the Consolidated Approach for Reporting Food Insecurity (CARI) calculated either using remote or face-to-face (F2F) modules with the standard CARI. The objective is to assess the comparability of these methodologies through group mean tests, specifically the student t-test.

The food security indicators analyzed include the following:

1. **Food Consumption Score (FCS)**
2. **Reduced Coping Strategies Index (rCSI)**
3. **Food Expenditure Score (FES)**
4. **Livelihood Coping Strategies**
5. **Income Sources and Income Change**

For each indicator, we conducted group mean tests to compare the results obtained from the different versions of CARI. This analysis allows us to evaluate the statistical differences between the remote and F2F modules compared to the standard CARI. The results of the group mean tests provide insights into the comparability of food security indicators derived from remote with F2F data obtained using the standard CARI methodology. Statistical significance levels of 95%, 99% and 99.9% were determined to assess the extent of differences between the methodologies.

Overall, this analysis aims to elucidate the extent to which food security assessments conducted remotely align with those conducted through traditional F2F interviews. These findings contribute to our understanding of the reliability and validity of remote data collection methods in assessing food security in hard to reach in emergency situations.

4.1 Sample Completion

Tables 1 & 2 show the total sample and number of household interviews completed by country and modality in the 3 pilot countries of the study respectively.

Table 1: Survey Completion by Country-F2F Sample

	Country			
	Myanmar	Ecuador	CAR	Total
N	833 (27.2%)	1,173 (38.3%)	1,057 (34.5%)	3,063 (100.0%)
Survey Progress				
Not Completed	223 (30.8%)	413 (57.0%)	88 (12.2%)	724 (23.6%)
Completed	610 (26.1%)	760 (32.5%)	969 (41.4%)	2,339 (76.4%)

As expected, the response rate in the F2F sample (Table 2) was higher than the remote (Table 1)in

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Table 2: Survey Completion by Country-Remote Sample

	Country			
	Myanmar	Ecuador	CAR	Total
N	832 (27.2%)	1,199 (39.3%)	1,023 (33.5%)	3,054 (100.0%)
Survey Progress				
Not Completed	349 (30.8%)	543 (57.0%)	747 (12.2%)	1,639 (53.7%)
Completed	483 (34.1%)	656 (46.4%)	276 (19.5%)	1,415 (46.3%)

all three countries. This was primarily due to the poor phone connection/network and not picking up the phone calls, and less than 1% of refusals by respondents. Nevertheless, this did not influence the sample balance for almost all the collected background information at the household level for Ecuador and Myanmar. In CAR, however, the study sample was sub-categorized into phone-owners and non-phone-owners to capture phone and network coverage differences. The balance test performed showed that the two samples were largely balanced over the two modalities. For more details about response rate, the final sample statistical balance analysis test (see attrition tables attached in Annex).

4.2 CARI vs. rCARI

Here, the standard CARI using FES from F2F surveys was compared to other ways of aggregating CARI (below and in Table 2):

Other ways of aggregating CARI (that are tested)

- rCARI-FES: Standard CARI using FES from remote surveys.
- CARI-Inc: CARI using income variable from F2F surveys.
- rCARI-Inc: Standard rCARI using income and income change variable from remote surveys (the CARI version that was used during COVID 19 Pandemic).

For the purposes of this interim analysis and report, results from CAR are restricted to only respondents that owned phones (the complete sample comprising those who did not own phones but received mobile phones from the study are also available).

Table 3: Lists the different aggregation methods and indicators used for each methodology

	Aggregation Methodology			
Indicator	**CARI (F2F)**	**CARI (F2F)**	**rCARI (Remote)**	**rCARI (Remote)**
FCS	Yes	Yes	Yes	Yes
FES	Yes	**No**	Yes	No
LCS	Yes	Yes	Yes	Yes
rCSI	Yes	Yes	Yes	Yes
Income source & income change	**No**	Yes	**No**	**Yes**

For more information on the calculation of the food security indicators included in each methodology

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refer to supporting documents attached in the Annex.

The first comparison is between rCARI-Inc (using the income and income change variable) with standard CARI presented in Table 3 for all of the pilot countries. In Myanmar and Ecuador, there was no statistically significant differences reported in the prevalence of food insecurity (moderately food insecure plus severely food insecure) between the two aggregation methodologies. However, in CAR, rCARI-Inc underestimated the prevalence of food insecurity by 21 percentage points compared to CARI-FES. Further, rCARI-Inc was identical to CARI-FES in estimating the percentage of moderately food insecure in Myanmar and Ecuador but underestimated.

In CAR, rCARI-Inc significantly underestimated the moderately food insecure numbers and severely food insecure by 15 and 6 percentage points, respectively, compared to CARI-FES.

Table 4: Impact of Survey Modality: Standard CARI-FES vs. Standard rCARI-Inc

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.08	0.07	-0.01	0.01	0.00	-0.01	0.01	0.01	-0.00
	(0.27)	(0.26)	(0.67)	(0.11)	(0.07)	(0.08)	(0.08)	(0.08)	(0.96)
Marginally Food Secure	0.44	0.41	-0.03	0.23	0.27	0.03	0.31	0.52	0.21***
	(0.50)	(0.49)	(0.42)	(0.42)	(0.44)	(0.13)	(0.46)	(0.50)	(0.00)
Moderately Food Insecure	0.44	0.49	0.05	0.59	0.62	0.03	0.60	0.45	-0.15**
	(0.50)	(0.50)	(0.23)	(0.49)	(0.48)	(0.22)	(0.49)	(0.50)	(0.00)
Severely Food Insecure	0.03	0.03	-0.01	0.16	0.11	-0.06**	0.08	0.02	-0.06**
	(0.18)	(0.16)	(0.59)	(0.37)	(0.31)	(0.00)	(0.27)	(0.14)	(0.00)
Food Insecure Group	0.47	0.51	0.04	0.76	0.73	-0.03	0.68	0.47	-0.21***
	(0.50)	(0.50)	(0.31)	(0.43)	(0.44)	(0.26)	(0.47)	(0.50)	(0.00)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 4 shows the results of rCARI calculated using the FES collected by phone and represents the scenario of collecting all CARI indicators over phone in remote interviews. As discussed earlier, the FES collected over the phone is a relatively shorter version of the standard FES that is usually collected in F2F surveys. This version collects the household's aggregate expenditures without going into the individual expenses. The results show that rCARI using FES consistently over-estimates the number of food insecure across all countries compared to the standard CARI by between 2 and 17 percentage points. This difference was statistically significant in Myanmar and Ecuador.

The CARI results calculated using the income and income change modules implemented by phone (rCARI-Inc) and in F2F interviews (CARI-Inc) are presented in Table 5. Results are different across the 3 pilot countries. The results show that rCARI-Inc significantly over-estimates the number of food insecure compared to CARI-Inc by between 9-10 percentage points in Myanmar and Ecuador.

Next, a comparison of CARI computed using FES is compared with income and income change variable modules asked to the same household/respondent in the same modality. This step allowed a comparison of these two indicators using the same respondents in the same modality. Results for the paired-sample test for the F2F modality are presented in Table 6, while that for the Remote modality is presented in Table 7. The main conclusion from Table 6 is that CARI computed using

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Table 5: Impact of Survey Modality: CARI-FES vs. rCARI-FES

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.08	0.03	-0.06***	0.01	0.00	-0.01**	0.01	0.01	-0.00
	(0.27)	(0.16)	(0.00)	(0.11)	(0.04)	(0.01)	(0.08)	(0.08)	(0.96)
Marginally Food Secure	0.44	0.34	-0.10**	0.23	0.07	-0.16***	0.31	0.29	-0.02
	(0.50)	(0.48)	(0.01)	(0.42)	(0.26)	(0.00)	(0.46)	(0.45)	(0.61)
Moderately Food Insecure	0.44	0.57	0.13***	0.59	0.69	0.10***	0.60	0.58	-0.02
	(0.50)	(0.50)	(0.00)	(0.49)	(0.46)	(0.00)	(0.49)	(0.50)	(0.65)
Severely Food Insecure	0.03	0.06	0.02	0.16	0.23	0.07**	0.08	0.13	0.05
	(0.18)	(0.23)	(0.17)	(0.37)	(0.42)	(0.00)	(0.27)	(0.33)	(0.14)
Food Insecure Group	0.47	0.63	0.16***	0.76	0.92	0.17***	0.68	0.71	0.02
	(0.50)	(0.48)	(0.00)	(0.43)	(0.27)	(0.00)	(0.47)	(0.46)	(0.61)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 6: Impact of Survey Modality: CARI-Inc vs. rCARI-Inc

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Secure	0.16	0.07	-0.09***	0.03	0.00	-0.02***	0.02	0.01	-0.01
	(0.37)	(0.26)	(0.00)	(0.17)	(0.07)	(0.00)	(0.14)	(0.08)	(0.18)
Marginally Food Secure	0.43	0.41	-0.02	0.33	0.27	-0.07**	0.51	0.52	0.01
	(0.50)	(0.49)	(0.68)	(0.47)	(0.44)	(0.01)	(0.50)	(0.50)	(0.86)
Moderately Food Insecure	0.39	0.49	0.10**	0.53	0.62	0.09***	0.45	0.45	0.01
	(0.49)	(0.50)	(0.01)	(0.50)	(0.48)	(0.00)	(0.50)	(0.50)	(0.89)
Severely Food Insecure	0.02	0.03	0.00	0.11	0.11	-0.00	0.02	0.02	-0.00
	(0.15)	(0.16)	(0.81)	(0.31)	(0.31)	(0.93)	(0.14)	(0.14)	(0.92)
Food Insecure Group	0.41	0.51	0.10**	0.64	0.73	0.09***	0.47	0.47	0.01
	(0.49)	(0.50)	(0.01)	(0.48)	(0.44)	(0.00)	(0.50)	(0.50)	(0.91)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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the income and income change variables significantly under reports the numbers of food insecure across all pilot countries (6% in Myanmar, 23% in Ecuador and 22% in CAR).

Table 7: FES vs Income Indicator: Paired CARI-FES vs. CARI-Inc in F2F Surveys

	Myanmar			Ecuador			CAR		
Food Security Category	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff	CARI-FES	CARI-Inc	Diff
Food Secure	0.08	0.16	0.08***	0.01	0.03	0.02	0.01	0.02	0.01*
	(0.01)	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)	(0.00)	(0.01)	(0.01)
Marginally Food Secure	0.44	0.43	-0.01	0.23	0.33	0.10***	0.31	0.51	-0.20***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Moderately Food Insecure	0.44	0.39	-0.05**	0.59	0.53	-0.06**	0.60	0.45	-0.16***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Severely Food Insecure	0.03	0.02	-0.01	0.16	0.11	-0.06***	0.08	0.02	-0.06***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)	(0.01)	(0.02)
Food Insecure Group	0.47	0.41	-0.06***	0.76	0.64	-0.12***	0.68	0.47	-0.22***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Observations	610	610	610	760	760	760	294	294	294

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
For CAR, the sample is limited to phone owners only.

The results of comparing FES and income indicator using same respondent households in the remote modality show similar trend of respondents significantly under-reporting their food insecurity using the income source and income change indicator by between 12 and 24 percentage points compared to the FES indicator (12% in Myanmar, 19% in Ecuador and 24% in CAR, see table 7).

Table 8: FES vs Income: Paired rCARI-FES vs. rCARI-Inc in Remote Surveys

	Myanmar			Ecuador			CAR		
Food Security Category	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff	rCARI-FES	rCARI-Inc	Diff
Food Secure	0.03	0.07	0.05**	0.00	0.00	0.00	0.01	0.01	0.00
	(0.01)	(0.02)	(0.02)	(0.00)	(0.00)	(0.00)	(0.01)	(0.01)	(0.00)
Marginally Food Secure	0.34	0.41	0.07*	0.07	0.27	0.19***	0.29	0.52	0.24***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)
Moderately Food Insecure	0.57	0.49	-0.09**	0.69	0.63	-0.07**	0.58	0.45	-0.13*
	(0.03)	(0.03)	(0.03)	(0.02)	(0.02)	(0.02)	(0.04)	(0.04)	(0.05)
Severely Food Insecure	0.06	0.03	-0.03*	0.23	0.11	-0.13***	0.18	0.02	-0.11***
	(0.02)	(0.01)	(0.01)	(0.02)	(0.01)	(0.01)	(0.03)	(0.01)	(0.03)
Food Insecure Group	0.63	0.51	-0.12***	0.92	0.73	-0.19***	0.71	0.47	-0.24***
	(0.03)	(0.03)	(0.03)	(0.01)	(0.02)	(0.02)	(0.04)	(0.04)	(0.04)
Observations	232	232	232	656	656	656	157	157	157

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
For CAR, the sample is limited to phone owners only.

The results in Tables 6 & 7, demonstrate that the income source and income change variable as formulated in its current form significantly under-reports food insecurity compared to using the traditional FES indicator.

4.3 Food Consumption Score (FCS)

The Food Consumption Score (FCS) is a composite score based on households' dietary diversity, food consumption frequency, and the relative nutritional value of different food groups. FCS is calculated using household food consumption of eight food groups during a 7-day reference period. The responses are aggregated, and weights applied to obtain a score, which is then used to categorize the households into one of three food consumption groups – acceptable, borderline, or poor food consumption.

The results of the FCS analysed across the three pilot countries are presented in Table 8. The

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results show that respondents tend to overstate their food security outcomes (report better food consumption) over the phone compared to F2F. A comparison of the FCS reported over the two modalities show statistical significance in Ecuador and CAR. However, there is no significant difference found in Myanmar. This means that more households are classified as having poor food consumption in the F2F survey compared to the remote.

Table 9: Impact of Survey Modality on Food Consumption Score and Groups

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Consumption Score	44.32	43.75	-0.57	46.28	52.54	6.25***	39.76	47.63	7.87***
	(12.36)	(12.15)	(0.45)	(21.57)	(24.12)	(0.00)	(17.58)	(17.45)	(0.00)
FCS: Acceptable	0.54	0.48	-0.06	0.54	0.62	0.09***	0.40	0.58	0.18***
	(0.50)	(0.50)	(0.07)	(0.50)	(0.49)	(0.00)	(0.49)	(0.50)	(0.00)
FCS: Borderline	0.40	0.45	0.05	0.24	0.22	-0.02	0.35	0.29	-0.06
	(0.49)	(0.50)	(0.09)	(0.43)	(0.42)	(0.35)	(0.48)	(0.45)	(0.19)
FCS: Poor	0.06	0.07	0.00	0.22	0.16	-0.07**	0.25	0.13	-0.12**
	(0.24)	(0.25)	(0.79)	(0.42)	(0.36)	(0.00)	(0.43)	(0.34)	(0.00)
Consumption over the past 7 days (cereals and tubers)	7.00	7.00	0.00	5.38	4.93	-0.45***	5.87	6.24	0.36*
	(0.12)	(0.00)	(0.32)	(2.28)	(2.32)	(0.00)	(1.86)	(1.33)	(0.02)
Consumption over the past 7 days (pulses)	1.63	1.82	0.20*	2.00	2.50	0.50***	3.74	3.74	-0.01
	(1.46)	(1.55)	(0.03)	(2.06)	(2.24)	(0.00)	(2.33)	(2.08)	(0.98)
Consumption over the past 7 days (dairy products)	0.17	0.18	0.01	1.38	2.03	0.65***	0.50	1.14	0.64**
	(0.67)	(0.62)	(0.79)	(2.02)	(2.32)	(0.00)	(1.57)	(2.27)	(0.00)
Consumption over the past 7 days (protein-rich foods)	3.22	3.01	-0.21	3.03	3.48	0.44***	1.72	2.09	0.37*
	(2.05)	(1.97)	(0.09)	(2.42)	(2.39)	(0.00)	(1.81)	(1.81)	(0.04)
Consumption over the past 7 days (vegetables)	6.42	6.48	0.06	3.93	4.75	0.82***	3.73	5.64	1.92***
	(1.36)	(1.21)	(0.43)	(2.82)	(2.56)	(0.00)	(2.93)	(1.66)	(0.00)
Consumption over the past 7 days (fruit)	1.73	1.43	-0.30**	2.58	2.90	0.31*	0.54	0.72	0.18
	(1.92)	(1.67)	(0.01)	(2.57)	(2.61)	(0.02)	(1.31)	(1.34)	(0.17)
Consumption over the past 7 days (oil)	6.67	6.37	-0.30***	5.38	5.40	0.02	4.90	5.79	0.89***
	(1.21)	(1.31)	(0.00)	(2.52)	(2.34)	(0.87)	(2.37)	(1.81)	(0.00)
Consumption over the past 7 days (sugar)	0.82	0.85	0.03	5.39	5.64	0.25*	2.36	3.53	1.17***
	(1.50)	(1.58)	(0.74)	(2.57)	(2.25)	(0.05)	(2.60)	(2.78)	(0.00)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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4.4 Reduced Coping Strategy Index (rCSI)

The reduced Coping Strategies index (rCSI) is used to compare households'stress levels due to food shortages. It measures the frequency and severity of the food consumption related behaviours by households to cope with food shortage within a 7 days recall period. The indicator is comprised of 5 standard strategies framed around how and who consumes food in times of shortage. It serves as a proxy indicator for food access.

A comparison of the coping strategy adoption based on F2F and remote interviews is reported in Table 9. The results show that in Myanmar and CAR, households tend to report adopting more coping strategies over the phone compared to F2F while, the opposite is found in Ecuador (where respondents are adopting significantly higher coping strategies in the F2F).

Table 10: Impact of Survey Modality on Reduced Coping Strategy Index Classification

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Reduced Coping Strategies Index (rCSI)	5.43	9.24	3.81***	19.86	18.50	-1.36*	14.35	17.27	2.92**
	(7.46)	(8.87)	(0.00)	(12.41)	(11.68)	(0.03)	(9.97)	(8.88)	(0.00)
rCSI: Low	0.59	0.31	-0.29***	0.08	0.06	-0.02	0.06	0.04	-0.02
	(0.49)	(0.46)	(0.00)	(0.28)	(0.24)	(0.09)	(0.25)	(0.21)	(0.36)
rCSI: Median	0.33	0.53	0.20***	0.44	0.51	0.08**	0.70	0.57	-0.14**
	(0.47)	(0.50)	(0.00)	(0.50)	(0.50)	(0.00)	(0.46)	(0.50)	(0.00)
rCSI: Severe	0.08	0.17	0.09***	0.48	0.43	-0.05	0.23	0.39	0.16***
	(0.26)	(0.37)	(0.00)	(0.50)	(0.50)	(0.05)	(0.42)	(0.49)	(0.00)
Relied on less preferred, less expensive food	1.92	2.86	0.94***	4.41	4.56	0.15	3.92	2.75	-1.17***
	(2.20)	(2.26)	(0.00)	(2.58)	(2.71)	(0.29)	(2.00)	(1.84)	(0.00)
Borrowed food or relied on help from friends or relatives	0.59	0.56	-0.03	1.64	1.75	0.11	0.77	0.89	0.12
	(1.20)	(1.13)	(0.69)	(1.95)	(2.13)	(0.33)	(1.15)	(1.25)	(0.32)
Reduced portion size of meals at meals time	0.56	0.74	0.18*	3.80	3.27	-0.53***	1.98	2.65	0.67**
	(1.32)	(1.42)	(0.03)	(2.70)	(2.85)	(0.00)	(2.27)	(2.14)	(0.00)
Restricted consumption by adults in order for young-children to eat	0.52	1.23	0.72***	1.87	1.51	-0.36**	1.73	2.70	0.97***
	(1.39)	(1.78)	(0.00)	(2.38)	(2.24)	(0.00)	(1.90)	(1.83)	(0.00)
Reduced the number of meals eaten per day	0.22	0.82	0.60***	2.77	2.66	-0.11	1.74	2.00	0.26
	(0.90)	(1.67)	(0.00)	(2.52)	(2.70)	(0.43)	(2.38)	(2.12)	(0.24)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

In terms of categorizing respondents into the 3 classes of rCSI, the remote surveys seem to classify more households into the severe category both in Myanmar and CAR than Ecuador. Although the direction of the difference is not consistent across the countries, the differences remain relatively minimal (ranging from -1.3 to +3.8 points) and since the rCSI has a low overall weight in the CARI aggregate, this minimal change in reporting between F2F and remote is expected have a negligible overall impact on the final CARI index.

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4.5 Livelihood Coping Strategy (LCS)

The livelihood coping strategy indicator is a household-level indicator that measures the household's capacity to deal with insufficient food consumption due to lack of food or lack of money to buy food in the medium- to long-term (30 days to 12 months) . The livelihood coping strategies are categorized into three severity levels: stress, crisis, and emergency. For the purposes of targeting and assistance, the LCS crisis and emergency categories are of relevance. Across the three pilot countries, in the F2F modality, households tended to report higher adoption of either no or stress-level livelihood coping strategies. However, the results are mixed comparing the adoption of crisis and emergency coping strategies, which are the two severity levels that are responsible for the moderately and severely food insecure CARI categories). While there is a significantly higher adoption of crisis and severe coping strategies by households during the remote interviews in Myanmar and CAR compared to F2F, the opposite was observed for crisis level in Ecuador. Worthy of mention is that in Ecuador, there was an unexplained over reporting of emergency-level coping severity (large 50 percentage points difference). This was driven by misreporting on the household engagement in illegal activities and begging over phone surveys. The over reporting of emergency coping led to the under classification of crisis coping in Ecuador (as the indicator takes the maximum severity of the coping strategies applied by the household). However, both crisis and emergency-level coping in Ecuador was still higher in the remote group than the F2F by 27 percentage points. Further investigation is needed to explain these results.

Table 11: Impact of Survey Modality on Livelihood Coping Strategy Index

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Maximum Livelihood Coping: None	0.26	0.12	-0.14***	0.09	0.02	-0.07***	0.20	0.11	-0.09*
	(0.44)	(0.32)	(0.00)	(0.29)	(0.14)	(0.00)	(0.40)	(0.32)	(0.01)
Maximum Livelihood Coping: Stress level	0.46	0.41	-0.05	0.28	0.08	-0.20***	0.61	0.43	-0.19***
	(0.50)	(0.49)	(0.10)	(0.45)	(0.28)	(0.00)	(0.49)	(0.50)	(0.00)
Maximum Livelihood Coping: Crisis level	0.25	0.33	0.08**	0.42	0.19	-0.23***	0.14	0.26	0.13**
	(0.43)	(0.47)	(0.00)	(0.49)	(0.39)	(0.00)	(0.34)	(0.44)	(0.00)
Maximum Livelihood Coping: Emergency level	0.03	0.14	0.10***	0.21	0.70	0.50***	0.05	0.20	0.15***
	(0.18)	(0.35)	(0.00)	(0.41)	(0.46)	(0.00)	(0.22)	(0.40)	(0.00)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

The implications of these results at this stage, though not conclusive, are that the LCS module as it is framed is difficult to ask over the phone. This might be addressed through increased enumerator training and simulation exercises or an overhaul of the LCS module to make it more practical and easier to administer over the phone. To better understand the LCS indicator, the individual strategies are further assessed across the pilot countries (see Tables 11-13).

Table 11 shows how the adoption of LCS strategies compare across the F2F and Remote samples in Myanmar. The results show a consistent pattern, where respondents reported significantly adopting m ore strategies in the remote survey (doing worse) compared to F2F sample in terms of how they cope with lack of food.

Table 12 shows how the adoption of LCS strategies compare across the F2F and Remote samples

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Table 12: Myanmar: Each Coping Strategy Used

Variable	F2F	Myanmar Remote	Diff
Used Stress coping: Sold household assets/goods	0.07	0.13	0.06**
	(0.26)	(0.33)	(0.00)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.53	0.67	0.14***
	(0.50)	(0.47)	(0.00)
Used Stress coping: Spent savings	0.24	0.32	0.08**
	(0.42)	(0.47)	(0.00)
Used Stress coping: Borrowed money	0.42	0.65	0.23***
	(0.49)	(0.48)	(0.00)
Used Crisis coping: Sold productive assets or means of transport	0.02	0.06	0.04**
	(0.14)	(0.23)	(0.00)
Used Crisis coping: Reduced expenses on health	0.25	0.40	0.15***
	(0.43)	(0.49)	(0.00)
Used Crisis coping: Withdrew children from school	0.01	0.06	0.05***
	(0.10)	(0.25)	(0.00)
Used Emergency coping: Send children to work for household income	0.01	0.05	0.04***
	(0.11)	(0.23)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.03	0.02*
	(0.11)	(0.18)	(0.02)
Used Emergency coping: Engaged in illegal income activities	0.02	0.07	0.05***
	(0.13)	(0.26)	(0.00)
Observations	610	483	1,093

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

in Ecuador. The result is that more households (> 50%) reported engaging in illegal activities to cope with a lack of food in the remote sample compared to the F2F. This however needs further investigation since it is unlikely that more than half of a population relied on begging and illegal activities to deal with food shortages.

The results in CAR (Table 13) are similar to Myanmar, where more households report adopting coping strategies significantly more often over the phone compared to F2F.

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Table 13: Ecuador: Each Coping Strategy Used

Variable	Ecuador F2F	Ecuador Remote	Ecuador Diff
Used Stress coping: Sold household assets/goods	0.10	0.15	0.05**
	(0.31)	(0.36)	(0.01)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.39	0.31	-0.08**
	(0.49)	(0.46)	(0.00)
Used Stress coping: Spent savings	0.62	0.65	0.03
	(0.49)	(0.48)	(0.18)
Used Stress coping: Borrowed money	0.64	0.64	0.01
	(0.48)	(0.48)	(0.80)
Used Crisis coping: Sold productive assets or means of transport	0.05	0.08	0.03
	(0.23)	(0.27)	(0.06)
Used Crisis coping: Reduced expenses on health and education	0.57	0.61	0.04
	(0.50)	(0.49)	(0.10)
Used Crisis coping: Withdrew children from school	0.08	0.11	0.03
	(0.27)	(0.31)	(0.07)
Used Emergency coping: Send children to work for household income	0.08	0.09	0.01
	(0.27)	(0.28)	(0.53)
Used Emergency coping: Begged and/or scavenged	0.10	0.23	0.13***
	(0.29)	(0.42)	(0.00)
Used Emergency coping: Engaged in illegal income activities	0.09	0.58	0.49***
	(0.28)	(0.49)	(0.00)
Observations	760	656	1,416

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 14: CAR: Each Coping Strategy Used

Variable	CAR F2F	CAR Remote	CAR Diff
Used Stress coping: Sold household assets/goods	0.28	0.46	0.18***
	(0.45)	(0.50)	(0.00)
Used Stress coping: Reduced expenses on health and education	0.23	0.43	0.20***
	(0.42)	(0.50)	(0.00)
Used Stress coping: Spent savings	0.47	0.50	0.03
	(0.50)	(0.50)	(0.50)
Used Stress coping: Borrowed money	0.46	0.53	0.07
	(0.50)	(0.50)	(0.18)
Used Crisis coping: Sold productive assets or means of transport	0.07	0.26	0.19***
	(0.26)	(0.44)	(0.00)
Used Crisis coping: Moved to somewhere else	0.05	0.14	0.09**
	(0.23)	(0.35)	(0.01)
Used Crisis coping: Withdrew children from school	0.03	0.05	0.02
	(0.17)	(0.22)	(0.32)
Used Emergency coping: Sold houses	0.03	0.14	0.11***
	(0.17)	(0.35)	(0.00)
Used Emergency coping: Begged and/or scavenged	0.01	0.07	0.06**
	(0.12)	(0.26)	(0.01)
Used Emergency coping: Engaged in illegal income activities	0.01	0.03	0.02
	(0.10)	(0.16)	(0.27)
Observations	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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4.6 Food Expenditure Share (FES)

The food expenditure share (FES) is the share of household's total consumption expenditures spent on food. It is a proxy food security indicator that is used to measure the economic vulnerability of a household. The higher the FES of a household, the more vulnerable/food insecure the household.

Table 14 presents the FES results as reported from the 3 pilot countries. In the first row, the FES shows that respondents consistently report higher FES over the phone compared to F2F e (Table 14). The difference, on average, is between 5-8 percentage points and is statistically significant across all countries.

Across all countries, in remote interviews, significantly more households are classified into FES >= 75%, which corresponds to severely food insecure – CARI) (Myanmar 48%, Ecuador 68% & CAR 54%). The differences for those who were classified as severely food insecure over the two modalities was larger (10 to 19 percentage points across all three-study countries).

Table 15: Impact of Survey Modality on Expenditure and Food Expenditure Share

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Food Expenditure Share	0.66	0.71	0.05***	0.71	0.79	0.08***	0.67	0.73	0.06*
	(0.15)	(0.14)	(0.00)	(0.19)	(0.16)	(0.00)	(0.20)	(0.24)	(0.01)
FES < 50 percent	0.13	0.08	-0.06*	0.15	0.05	-0.10***	0.20	0.16	-0.04
	(0.34)	(0.27)	(0.01)	(0.36)	(0.22)	(0.00)	(0.40)	(0.37)	(0.27)
FES: 50-65 percent	0.30	0.22	-0.08*	0.17	0.09	-0.08***	0.23	0.17	-0.06
	(0.46)	(0.41)	(0.01)	(0.38)	(0.29)	(0.00)	(0.42)	(0.38)	(0.15)
FES: 65-75 percent	0.27	0.26	-0.01	0.19	0.18	-0.01	0.13	0.13	-0.01
	(0.44)	(0.44)	(0.76)	(0.39)	(0.38)	(0.54)	(0.34)	(0.33)	(0.87)
FES: >= 75 percent	0.30	0.45	0.15***	0.49	0.68	0.19***	0.44	0.54	0.10*
	(0.46)	(0.50)	(0.00)	(0.50)	(0.47)	(0.00)	(0.50)	(0.50)	(0.04)
Observations	610	232	842	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
In Myanmar, the Short Form in Remote Survey was dropped due to translation issues.

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4.7 Income Source and Income Change

The income indicator captures the household income source and income change experienced in the last 12 months prior to the interview. This indicator is a recent addition to the remote CARI console to reflect the economic situation of the household in the absence of ECMEN (recommended) or the FES. This was especially important during the COVID-19 pandemic when time did not allow for the collection of expenditure data to calculate FES. The main type of income generating activity and how the economic situation at the time impacted income either positively or negatively was considered as a proxy for household vulnerability. Based on the regularity of income (regular/irregular) and the change to income (no change/increasing/decreasing) the household is classified into one of 4 CARI categories (see Table 15), with the 'irregular & reduced income', and 'no income sources' categories corresponding to FES 65-75% and >=75% categories, respectively.

The results of the income and income change indicator compared by collection modality for the three pilot countries are presented in Table 15. The results show that more households in Myanmar and Ecuador underestimated their food insecurity when interviewed over the phone compared to F2F, by 1% in Myanmar and 7% in Ecuador. The opposite effect was found in CAR, where remote interviews resulted in respondents overestimated their food insecurity by 11 percentage points.

Table 16: Impact of Survey Modality on Income

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote	Diff	F2F	Remote	Diff	F2F	Remote	Diff
Income: Regular and Unchanged/Increasing	0.22	0.19	-0.03	0.19	0.25	0.06**	0.45	0.26	-0.19***
	(0.41)	(0.39)	(0.30)	(0.40)	(0.44)	(0.01)	(0.50)	(0.44)	(0.00)
Income: Regular and Reduced OR Irregular and Unchanged/Increasing	0.39	0.41	0.03	0.38	0.41	0.03	0.42	0.50	0.08
	(0.49)	(0.49)	(0.36)	(0.49)	(0.49)	(0.25)	(0.49)	(0.50)	(0.13)
Income: Irregular and Reduced	0.28	0.33	0.04	0.26	0.24	-0.01	0.12	0.23	0.11**
	(0.45)	(0.47)	(0.11)	(0.44)	(0.43)	(0.54)	(0.32)	(0.42)	(0.00)
Income: No income sources	0.11	0.07	-0.05**	0.17	0.09	-0.08***	0.01	0.01	-0.00
	(0.32)	(0.25)	(0.01)	(0.37)	(0.29)	(0.00)	(0.12)	(0.11)	(0.94)
Observations	610	483	1,093	760	656	1,416	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

As discussed earlier, the income source and income change indicator is a relatively new food security proxy that is being piloted. Further research needs to be conducted on the potential use of the main income source and income change on how the employment types may drive the indicator. This is important given that employment types and their economic importance could be country-specific; thus, applying a generalized global classification may result in misclassifying household food insecurity. Also, to properly situate the performance of this variable against the FES, more analysis needs to be performed beyond mean comparisons.

In the next section, the analysis and results will focus on the comparison of the different versions of the indicators. The level of comparison is the length of the forms (long- vs. short version) by data collection modality (F2F and Remote).

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5 Module Length and Food Security Indicator Reporting

As a secondary objective, the study also piloted a number of different versions of the assessment indicators (shorter than the standard versions). The reason was to see if shortened versions use could be recommended in phone interviews when time/length of interview has been shown to be correlated with data quality. Specifically, FCS, FES, LCS-EN, and LCS-FS modules will be compared and tested. This would allow recommendation for the use of the shorter version of the FES module over the phone or in F2F surveys to reduce interviewee fatigue. The results (comparison of short and long (standard module length) is to serve as a guide to revising the current modules (long) if need be, especially if the shorter versions of each module produces numbers that statistically comparable to the long versions.

5.1 Standard FCS module vs. Short FCS module

In this section, the results are compared between the standard FCS module (based on 16 questions to ask main food groups & subgroups) with a collapsed shorter FCS module (that relies on 9 questions to ask about the consumption of only the main food groups).

In Table 16, the first comparison module length is done for the F2F sample for all pilot countries. In terms of the two version of FCS, the aggregated FCS results show that in Myanmar and Ecuador (except for CAR where the short version significantly overstates food consumption by 4.35 points), there are no significant difference between the short and long FCS. This means that the shorter FCS module in CAR would minimally under-report number of food insecure households. However, when looking at the disaggregated FCS by the 3 consumption groups, the poor group (corresponding to severe food insecure category of CARI), show no statistical difference across all 3 pilot countries.

5.2 Standard Expenditure module vs. Short Expenditure module

This section of the analysis tested whether expenditures could be measured through administering fewer questions. Four different versions of the expenditure module (2 versions for the F2F modality & the 2 versions for the remote modality) differentiated by the number of questions it entails were carved and tested.

- **F2F Modality**
    1. Standard expenditure module that contains all standard household expenditure items (a total of 154 questions comprising of food and non-food expenditure).
    2. Short expenditure module (made up of 108 collapsed questions covering the household expenditures for food and non-food items).
- **Remote Modality**
    1. Shorter expenditure module (made up of 23 aggregated questions each for food and

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Table 17: Impact of Module Length on FCS Reporting

	Myanmar F2F			Ecuador F2F			CAR F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
Food Consumption Score	44.67	43.96	-0.71	46.42	46.16	-0.26	37.57	41.92	4.35*
	(12.47)	(12.27)	(0.48)	(21.61)	(21.55)	(0.87)	(16.98)	(17.95)	(0.03)
FCS: Acceptable	0.54	0.53	-0.02	0.54	0.53	-0.02	0.32	0.48	0.16**
	(0.50)	(0.50)	(0.69)	(0.50)	(0.50)	(0.65)	(0.47)	(0.50)	(0.01)
FCS: Borderline	0.38	0.42	0.04	0.23	0.26	0.03	0.42	0.28	-0.14*
	(0.49)	(0.49)	(0.28)	(0.42)	(0.44)	(0.29)	(0.49)	(0.45)	(0.01)
FCS: Poor	0.08	0.05	-0.03	0.23	0.21	-0.02	0.26	0.24	-0.02
	(0.26)	(0.22)	(0.18)	(0.42)	(0.41)	(0.58)	(0.44)	(0.43)	(0.74)
Consumption over the past 7 days (cereals and tubers)	6.99	7.00	0.01	5.38	5.38	-0.00	5.79	5.95	0.17
	(0.17)	(0.00)	(0.32)	(2.31)	(2.25)	(1.00)	(1.96)	(1.75)	(0.45)
Consumption over the past 7 days (pulses)	1.60	1.66	0.06	1.91	2.09	0.18	3.71	3.78	0.08
	(1.42)	(1.50)	(0.62)	(2.07)	(2.05)	(0.23)	(2.21)	(2.45)	(0.77)
Consumption over the past 7 days (dairy products)	0.16	0.17	0.01	1.35	1.40	0.06	0.55	0.45	-0.10
	(0.77)	(0.56)	(0.86)	(1.96)	(2.08)	(0.70)	(1.61)	(1.53)	(0.60)
Consumption over the past 7 days (protein-rich foods)	3.35	3.10	-0.25	3.06	3.00	-0.06	1.66	1.78	0.11
	(2.04)	(2.04)	(0.13)	(2.45)	(2.39)	(0.73)	(1.69)	(1.92)	(0.59)
Consumption over the past 7 days (vegetables)	6.37	6.47	0.10	4.00	3.87	-0.13	2.10	5.34	3.24***
	(1.40)	(1.32)	(0.36)	(2.79)	(2.86)	(0.53)	(2.76)	(2.08)	(0.00)
Consumption over the past 7 days (fruit)	1.76	1.69	-0.07	2.77	2.40	-0.38*	0.46	0.62	0.16
	(1.97)	(1.88)	(0.64)	(2.59)	(2.53)	(0.04)	(1.26)	(1.36)	(0.29)
Consumption over the past 7 days (oil)	6.66	6.68	0.02	5.59	5.17	-0.42*	4.69	5.10	0.41
	(1.21)	(1.21)	(0.84)	(2.42)	(2.59)	(0.02)	(2.39)	(2.34)	(0.14)
Consumption over the past 7 days (sugar)	0.81	0.83	0.03	5.45	5.32	-0.14	2.25	2.47	0.22
	(1.58)	(1.41)	(0.83)	(2.55)	(2.59)	(0.47)	(2.39)	(2.79)	(0.47)
Observations	305	305	610	377	383	760	146	148	294

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

non-food expenditure respectively).

2. Extremely short expenditure module (3 collapsed food and 2 non-food expenditure questions respectively).

The food expenditure amount always covered a 7 day while non-food covers a 1 & 6 months except in the very short expenditure in remote modality where it covers 1 month. First, the results of the F2F modules are compared and presented in Table 17. The FES remains stable in the overall aggregates between long and short modules. The statistical test between the long and short FES modules implemented in the F2F interviews show no significant difference across all 3 pilot countries. Although the absolute expenditure amount for the collapsed/short module is always lower across pilot countries, the difference is not statistically significant.

Table 18: Impact of Module Length on Expenditure and FES in F2F Surveys

	Myanmar F2F			Ecuador F2F			CAR F2F		
Variable	Long	Short	Diff	Long	Short	Diff	Long	Short	Diff
Food Expenditure Share	0.66	0.66	0.00	0.72	0.70	-0.02	0.67	0.67	0.00
	(0.15)	(0.14)	(0.89)	(0.19)	(0.18)	(0.14)	(0.19)	(0.20)	(0.92)
FES < 50 percent	0.15	0.12	-0.03	0.14	0.16	0.03	0.20	0.20	0.00
	(0.36)	(0.32)	(0.24)	(0.34)	(0.37)	(0.30)	(0.40)	(0.40)	(0.93)
FES: 50-65 percent	0.29	0.31	0.03	0.18	0.17	-0.00	0.23	0.22	-0.01
	(0.45)	(0.46)	(0.48)	(0.38)	(0.38)	(1.00)	(0.42)	(0.42)	(0.84)
FES: 65-75 percent	0.26	0.28	0.02	0.17	0.21	0.04	0.15	0.11	-0.04
	(0.44)	(0.45)	(0.58)	(0.38)	(0.41)	(0.17)	(0.36)	(0.32)	(0.37)
FES: >= 75 percent	0.30	0.29	-0.01	0.52	0.45	-0.07	0.42	0.46	0.04
	(0.46)	(0.46)	(0.72)	(0.50)	(0.50)	(0.07)	(0.49)	(0.50)	(0.47)
Observations	305	305	610	377	383	760	146	148	294

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Comparing the two remote versions of the FES module Table 18, (excluding Myanmar data be-

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cause of a translation error that was detected in the questionnaire after data collection) show that the short remote version produced slightly higher absolute FES in both countries; however, the difference was only statistically significant in CAR (12 percentage points).

Table 19: Impact of Module Length on Expenditure and FES in Remote Surveys

	Ecuador RM			CAR RM		
Variable	Long	Short	Diff	Long	Short	Diff
Food Expenditure Share	0.78	0.80	0.02	0.65	0.78	0.12**
	(0.15)	(0.17)	(0.06)	(0.26)	(0.21)	(0.00)
FES < 50 percent	0.04	0.06	0.02	0.22	0.12	-0.10
	(0.20)	(0.24)	(0.33)	(0.41)	(0.33)	(0.12)
FES: 50-65 percent	0.08	0.10	0.02	0.18	0.16	-0.02
	(0.28)	(0.30)	(0.40)	(0.39)	(0.37)	(0.73)
FES: 65-75 percent	0.23	0.13	-0.10***	0.18	0.09	-0.10
	(0.42)	(0.33)	(0.00)	(0.39)	(0.28)	(0.09)
FES: >= 75 percent	0.65	0.71	0.07	0.42	0.63	0.22**
	(0.48)	(0.45)	(0.07)	(0.50)	(0.49)	(0.01)
Observations	324	332	656	65	92	157

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
Myanmar is excluded due to translation errors.

Third, a comparison across modality where the two remote versions are compared with the standard F2F expenditure module (all the expenditure questions). The results of these are presented in Tables 19 & 20. Table 19 presents the results of the comparison between the long remote version with the standard F2F expenditure module for all 3 pilot countries. A number of findings can be deduced from Table 19. First, aggregate FES was significantly higher in the long remote module than the standard F2F FES module in Myanmar (5%) and Ecuador (7%). In CAR, however, it was lower, but the difference was not statistically significant. Second, disaggregating the absolute FES into the 4 CARI FES categories shows that the long version of the remote expenditure module classified significantly more households as severely food insecure (FES >= 75), compared to the standard, with 15 percentage more in Myanmar, & 16 percentage in Ecuador.

Table 20: Impact of Module Length on Expenditure and FES: F2F vs. Remote Long

	Myanmar			Ecuador			CAR		
Variable	F2F	Remote Long	Diff	F2F	Remote Long	Diff	F2F	Remote Long	Diff
Food Expenditure Share	0.66	0.71	0.05***	0.71	0.78	0.07***	0.67	0.65	-0.02
	(0.15)	(0.14)	(0.00)	(0.19)	(0.15)	(0.00)	(0.20)	(0.26)	(0.66)
FES < 50 percent	0.13	0.08	-0.06*	0.15	0.04	-0.11***	0.20	0.22	0.01
	(0.34)	(0.27)	(0.01)	(0.36)	(0.20)	(0.00)	(0.40)	(0.41)	(0.79)
FES: 50-65 percent	0.30	0.21	-0.08*	0.17	0.08	-0.09***	0.23	0.18	-0.04
	(0.46)	(0.41)	(0.01)	(0.38)	(0.28)	(0.00)	(0.42)	(0.39)	(0.42)
FES: 65-75 percent	0.27	0.26	-0.01	0.19	0.23	0.04	0.13	0.18	0.05
	(0.44)	(0.44)	(0.84)	(0.39)	(0.42)	(0.15)	(0.34)	(0.39)	(0.32)
FES: >= 75 percent	0.30	0.45	0.15***	0.49	0.65	0.16***	0.44	0.42	-0.02
	(0.46)	(0.50)	(0.00)	(0.50)	(0.48)	(0.00)	(0.50)	(0.50)	(0.73)
Observations	610	233	843	760	324	1,084	294	65	359

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Comparing the results of the extremely short remote module (5 questions) with the standard expenditure module (excluding Myanmar data) shows that the extremely short module significantly inflates the FES in aggregate terms across the two countries (Ecuador, 9%, CAR, 11%). In terms of

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food insecurity classification, using the extremely short module significantly classifies more households (Ecuador, 22% and CAR, 19%) into FES category 4 (FES >= 75%) than the standard expenditure module.

Table 21: Impact of Module Length on Expenditure and FES: F2F vs. Remote Short

	Ecuador			CAR		
Variable	F2F	Remote Short	Diff	F2F	Remote Short	Diff
Food Expenditure Share	0.71	0.80	0.09***	0.67	0.78	0.11***
	(0.19)	(0.17)	(0.00)	(0.20)	(0.21)	(0.00)
FES < 50 percent	0.15	0.06	-0.09***	0.20	0.12	-0.08*
	(0.36)	(0.24)	(0.00)	(0.40)	(0.33)	(0.05)
FES: 50-65 percent	0.17	0.10	-0.07***	0.23	0.16	-0.06
	(0.38)	(0.30)	(0.00)	(0.42)	(0.37)	(0.16)
FES: 65-75 percent	0.19	0.13	-0.06**	0.13	0.09	-0.05
	(0.39)	(0.33)	(0.01)	(0.34)	(0.28)	(0.20)
FES: >= 75 percent	0.49	0.71	0.22***	0.44	0.63	0.19**
	(0.50)	(0.45)	(0.00)	(0.50)	(0.49)	(0.00)
Observations	760	332	1,092	294	92	386

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
Myanmar is excluded due to translation errors.

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6 LCS for Food Security vs LCS for Essential Needs

In food security assessments, two versions of the livelihood coping strategies module can be administered depending on the aims of the assessment. While the LCS-FS (food security) is used when the main objective of the assessment is on food security outcomes, the LCS-EN (essential needs) version is used when the scope of assessment broader than food security to include essential needs like shelter, health, education, etc. It is important to note that in this study, the LCS-FS consisted of 10 questions on 10 livelihood coping strategies selected for the context (comprised of 4 stress, 3 crisis and 3 emergency coping strategies), while the LCS-EN module consisted of the same formulation but was slightly longer with 3 additional questions asking about the main reason for applying the coping strategies indicated by stress level.

The objective here was to assess whether LCS-FS and LCS-EN were comparable (using the conversion). The two versions of the LCS module were implemented only in the F2F modality, hence comparison is only in this sub-sample. Table 21 shows the computed LCS comparison by version across pilot countries while Tables 22-24 presents the comparison of the different LCS-FCS by country.

6.1 Aggregate LCS-FS/EN (Maximum livelihood coping strategies)

A comparison of the severity of the LCS applied (see Table 21) shows that across pilot countries, there were no statistically significant differences between the proportion of households adopting LCS strategies to meet their food security or essential needs.

Table 22: Myanmar: LCS for Essential Needs vs. Food Security

	Myanmar F2F			Ecuador F2F			CAR F2F		
Variable	EN	FS	Diff	EN	FS	Diff	EN	FS	Diff
Maximum Livelihood Coping: None	0.25	0.26	0.01	0.11	0.07	-0.04	0.19	0.21	0.02
	(0.44)	(0.44)	(0.78)	(0.32)	(0.26)	(0.05)	(0.40)	(0.41)	(0.71)
Maximum Livelihood Coping: Stress level	0.47	0.45	-0.02	0.26	0.30	0.05	0.60	0.62	0.02
	(0.50)	(0.50)	(0.63)	(0.44)	(0.46)	(0.16)	(0.49)	(0.49)	(0.74)
Maximum Livelihood Coping: Crisis level	0.26	0.24	-0.01	0.45	0.40	-0.05	0.14	0.13	-0.02
	(0.44)	(0.43)	(0.71)	(0.50)	(0.49)	(0.15)	(0.35)	(0.34)	(0.70)
Maximum Livelihood Coping: Emergency level	0.02	0.05	0.02	0.18	0.23	0.05	0.06	0.04	-0.02
	(0.15)	(0.21)	(0.12)	(0.39)	(0.42)	(0.11)	(0.24)	(0.20)	(0.41)
Observations	305	305	610	377	383	760	146	148	294

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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6.2 Specific Coping Strategies Adoption by Country

Here, the individual livelihood coping strategies by country are compared (the LCS indicator is usually context specific and thus may be different strategies across countries) between the two modules (LCS-FS vs. LCS-EN). Table 22 shows the results for Myanmar, Table 23 Ecuador, and Table 24 for CAR, respectively.

Table 23: Myanmar: LCS for Essential Needs vs. Food Security

	Myanmar F2F		
Variable	LCS EN	LCS FS	Diff
Used Stress coping: Sold household assets/goods	0.07	0.08	0.01
	(0.25)	(0.27)	(0.62)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.54	0.53	-0.01
	(0.50)	(0.50)	(0.81)
Used Stress coping: Spent savings	0.22	0.25	0.02
	(0.42)	(0.43)	(0.48)
Used Stress coping: Borrowed money	0.46	0.38	-0.08*
	(0.50)	(0.49)	(0.04)
Used Crisis coping: Sold productive assets or means of transport	0.02	0.02	-0.00
	(0.15)	(0.14)	(0.79)
Used Crisis coping: Reduced expenses on health	0.26	0.25	-0.01
	(0.44)	(0.43)	(0.82)
Used Crisis coping: Withdrew children from school	0.01	0.01	0.00
	(0.10)	(0.10)	(0.99)
Used Emergency coping: Send children to work for household income	0.00	0.02	0.02
	(0.06)	(0.14)	(0.06)
Used Emergency coping: Begged and/or scavenged	0.01	0.01	-0.00
	(0.11)	(0.10)	(0.71)
Used Emergency coping: Engaged in illegal income activities	0.01	0.02	0.01
	(0.10)	(0.15)	(0.20)
Observations	305	305	610

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

To conclude, as far as the LCS analysis from the 3 pilot countries is concerned, there is no statistical difference between LCS-FS and LCS-EN.

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Table 24: Ecuador: LCS for Essential Needs vs. Food Security

	Ecuador F2F		
Variable	LCS EN	LCS FS	Diff
Used Stress coping: Sold household assets/goods	0.10	0.11	0.01
	(0.30)	(0.31)	(0.60)
Used Stress coping: Purchased food/non-food on credit (incur debts)	0.37	0.41	0.04
	(0.48)	(0.49)	(0.22)
Used Stress coping: Spent savings	0.61	0.63	0.01
	(0.49)	(0.48)	(0.69)
Used Stress coping: Borrowed money	0.62	0.66	0.04
	(0.49)	(0.48)	(0.22)
Used Crisis coping: Sold productive assets or means of transport	0.06	0.05	-0.01
	(0.23)	(0.22)	(0.59)
Used Crisis coping: Reduced expenses on health and education	0.55	0.58	0.04
	(0.50)	(0.49)	(0.32)
Used Crisis coping: Withdrew children from school	0.09	0.07	-0.02
	(0.29)	(0.25)	(0.20)
Used Emergency coping: Send children to work for household income	0.07	0.08	0.01
	(0.26)	(0.27)	(0.73)
Used Emergency coping: Begged and/or scavenged	0.07	0.12	0.05*
	(0.25)	(0.33)	(0.01)
Used Emergency coping: Engaged in illegal income activities	0.08	0.09	0.00
	(0.28)	(0.28)	(0.95)
Observations	377	383	760

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 25: CAR: LCS for Essential Needs vs. Food Security

	CAR F2F		
Variable	LCS EN	LCS FS	Diff
Used Stress coping: Sold household assets/goods	0.32	0.24	-0.08
	(0.47)	(0.43)	(0.14)
Used Stress coping: Reduced expenses on health and education	0.24	0.22	-0.02
	(0.43)	(0.41)	(0.63)
Used Stress coping: Spent savings	0.45	0.49	0.05
	(0.50)	(0.50)	(0.41)
Used Stress coping: Borrowed money	0.51	0.42	-0.09
	(0.50)	(0.50)	(0.13)
Used Crisis coping: Sold productive assets or means of transport	0.09	0.06	-0.03
	(0.29)	(0.24)	(0.36)
Used Crisis coping: Moved to somewhere else	0.07	0.04	-0.03
	(0.25)	(0.20)	(0.29)
Used Crisis coping: Withdrew children from school	0.03	0.03	0.01
	(0.16)	(0.18)	(0.75)
Used Emergency coping: Sold houses	0.05	0.01	-0.03
	(0.21)	(0.12)	(0.09)
Used Emergency coping: Begged and/or scavenged	0.01	0.02	0.01
	(0.08)	(0.14)	(0.32)
Used Emergency coping: Engaged in illegal income activities	0.01	0.01	0.01
	(0.08)	(0.12)	(0.57)
Observations	146	148	294

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

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7 Summary and conclusions

Disclaimer: the following conclusions are preliminary based on the data collected from the initial 3 pilot countries and thus should be interpreted with consideration to the following limitations of the study. The results from CAR are based on a sub-sample of the overall study sample (only phone owners'). Also, in Myanmar, results of FES and rCARI are based on half of the original households assigned to receive the remote interviews because of an error in translation that affected the expenditure module.

From the preliminary analysis of the available data, the following initially conclusions are made:

- rCARI calculated using the income & income change indicator (rCARI-Inc) is not statistically different from the standard/conventional CARI (CARI-FES) when used in Myanmar & Ecuador. In CAR, however, where it was significantly different, rCARI-Inc tended to underestimate food insecurity by 21 percentage points.
- Analysing the same households interviewed through the F2F modality revealed that CARI-Inc led to a lower food insecurity estimation by 6 to 22 percentage points compared to CARI-FES.
- FCS, rCSI, FES and income sources indicators collected by phone tended to give similar or slightly different results compared to F2F data collection, with differences in FCS (0 to 8 points), rCSI (-1 to 4 points) FES (5-8 points), and income sources (0 to 8 percentage points in category 4) (1 to 11 percentage points in category 3).
- Regardless of the collection modality, the absolute prevalence of the FES indicator (looking at category 3 and 4, i.e., FES > 65% and FES >75%, respectively) was much higher than the absolute prevalence of the income source indicator (looking at the irregular income + no income categories). The difference between the income source and FES prevalence from F2F was 18 percentage points in Myanmar, 25 percentage points in Ecuador and 44 percentage points in CAR.
- LCS collected by phone tended to produce a higher estimation of food insecurity than those collected through F2F. The difference seems to have mainly been driven by the LCS crisis + emergency coping (18-28 percentage points).
- Although LCS overestimated food insecurity in the 3 pilot countries between 18% and 28% points, the conservative lower estimate of the income indicator compared to FES (-18% to -44% points) could explain the equivalence between standard CARI (CARI-FES) with the (rCARI-Inc) in Myanmar and Ecuador (Table 3) and the underestimation of food insecurity in CAR (-44 percentage points).

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7.1 Next Steps...

1. Resume data collection in the remaining pilot study countries.
2. Conduct further in-depth analysis of the data - exploring the association between the indicators especially FES and Income indicators and perform other statistical diagnostic analysis.
3. Aggregate the full sample from CAR.
4. Apply weights to address the issues of lower response rates in the case of CAR and in other pilot countries where practicable.
5. Prepare the initial manuscripts for scientific publication.

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8 Challenges of the study by country

In this section, the main challenges faced in the 3 pilot countries are listed by country.

8.1 Myanmar

1. Outdated household listing data
2. Poor/unstable network coverage
3. Mobility of respondents/households
4. Absence of baseline data (for assessing the attrition characteristics)
5. Limited coverage of the study population (covered only IDPs)
6. Typhoon season affected the implementation period.
7. Restricted access to the field
8. Distrust from respondents

8.2 Ecuador

1. Issues of security for enumerators in some communities.
    a. Parts of the country were not accessible due to safety concerns for the face-to-face enumerators.
2. Misunderstanding with the local authorities preventing field access by F2F team.
    a. Dues to political reasons, some local authorities refused enumerators access leading to dropping of these communities.
    b. Distrust from local authorities and in some cases respondents.
3. Invalid respondent of household contact information (collected during the listing phase)
4. Poor phone coverage in remote areas
5. Disproportionate ownership of phones between urban and rural areas (low phone ownership in rural areas)
6. Low response rate particularly among the remote sample
    a. High attrition between the time of receiving consent and actual interview date.

8.3 CAR

1. Restricted access to the field because of political situation at the time,

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2. Bad phone network/infrastructure
    a. Most parts of the country do not have phone network coverage.
    b. In places where there is coverage, poor infrastructure and frequent electricity cuts affected call quality.
    c. No access to electricity for charging phones particularly in rural areas.
3. Low phone ownership in the population
    a. Only 25% of households had phones (similar to the national statistic)
4. Logistical challenges
    a. Relied on UNHAS (United Nations Humanitarian Air Service) for enumerator deployment. This affected the timely start of the exercise (flights to study areas twice a week)
    b. Acquisition and distribution of phones/SIM cards-Challenges with moving them to the field for distribution.
5. Low response rate in the remote modality (High Attrition)
6. Lack of an established data collection platform to collect phone survey-ICASEES
7. Weather disruptions/rainy season

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10 Annexes

10.1 Balance tables based on observable characteristics

Table 26: Myanmar Sample Balance Table (completed survey)

Variable	F2F	Myanmar Remote	Diff
Respondent Age	42.24	38.26	-3.98***
	(13.41)	(13.58)	(0.00)
Female Respondent	0.81	0.80	-0.01
	(0.39)	(0.40)	(0.61)
Head of Household Respondent	0.42	0.35	-0.07*
	(0.49)	(0.48)	(0.02)
Household Size	5.52	5.49	-0.03
	(2.20)	(2.18)	(0.84)
Number of household members contributing to income	1.41	1.41	-0.00
	(0.85)	(0.82)	(0.98)
Number of household income sources	1.18	1.24	0.07
	(0.68)	(0.59)	(0.09)
Observations	610	483	1,093

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.

Table 27: Ecuador Sample Balance Table (completed survey)

Variable	F2F	Ecuador Remote	Diff
Respondent Age	46.19	43.00	-3.19**
	(17.97)	(16.84)	(0.00)
Female Respondent	0.77	0.76	-0.01
	(0.42)	(0.43)	(0.51)
Head of Household Respondent	0.61	0.53	-0.08**
	(0.49)	(0.50)	(0.00)
Household Size	4.37	4.57	0.19
	(2.24)	(2.35)	(0.11)
Number of household members contributing to income	1.47	1.45	-0.01
	(0.84)	(0.80)	(0.75)
Number of household income sources	1.40	1.50	0.11**
	(0.72)	(0.80)	(0.01)
Observations	760	656	1,416

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
Figures might be changed slightly after roster inconsistency corrections.

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Table 28: CAR Sample Balance Table (completed surveys)

Variable	F2F	CAR Remote	Diff
Respondent Age	37.80	37.70	-0.10
	(13.66)	(12.39)	(0.91)
Female Respondent	0.54	0.43	-0.11**
	(0.50)	(0.50)	(0.00)
Head of Household Respondent	0.64	0.70	0.06
	(0.48)	(0.46)	(0.07)
Household Size	6.14	7.73	1.59***
	(2.59)	(3.12)	(0.00)
Number of household members contributing to income	1.76	1.66	-0.10
	(0.93)	(0.77)	(0.08)
Number of household income sources	1.50	1.46	-0.03
	(0.65)	(0.63)	(0.44)
Observations	969	276	1,245

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
Household listing information was used in this table.

Table 29: CAR Sample Balance Table (Phone owner subsample)

Variable	F2F	CAR Remote	Diff
List respondent age	36.27	39.03	2.76*
	(12.55)	(12.38)	(0.03)
Household head is the respondent	0.72	0.78	0.05
	(0.45)	(0.42)	(0.21)
Female Household Head	0.19	0.25	0.06
	(0.39)	(0.44)	(0.12)
Married Household Head	0.27	0.27	0.01
	(0.44)	(0.45)	(0.85)
Household head years of education	7.55	7.92	0.38
	(4.13)	(4.69)	(0.40)
Displaced household	0.09	0.15	0.06
	(0.29)	(0.36)	(0.07)
Household size	7.72	8.39	0.67*
	(3.44)	(3.40)	(0.05)
Number of household member younger than 15	4.04	4.31	0.27
	(2.27)	(2.03)	(0.20)
Number of household member older than 60	0.21	0.21	-0.00
	(0.49)	(0.44)	(0.99)
Has household member with disability/chronic disease	0.30	0.31	0.01
	(0.46)	(0.46)	(0.83)
Number of rooms	2.89	3.13	0.24
	(1.24)	(1.27)	(0.05)
Household Crowding Index	3.02	2.97	-0.05
	(1.70)	(1.26)	(0.73)
Household has impoverished roof	0.68	0.63	-0.05
	(0.47)	(0.48)	(0.33)
Household has impoverished floor	0.90	0.92	0.02
	(0.30)	(0.28)	(0.50)
Observations	294	157	451

Note: Robust standard errors in parentheses; * p<0.05, ** p<0.01, *** p<0.001.
Household listing information was used in this table.

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10.2 Supporting documents

10.2.1 CARI resources

- CARI methodology Consolidated Approach for Reporting Indicators of Food Security (CARI) Guidelines - Third edition
- FCS
- rCSI
- LCS
- FES

10.2.2 Calling protocol

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rCARI Validation Remote Survey Calling Protocol

Minimum call attempts and time coverage

- The respondent will be called as many times as needed (up to **20 times or more** over **several days (a maximum of 6 working days)**

Call time (days and hours)

- We shall be conducting surveys across working and non-working hours as stated in the work arrangement section.
- We are trying to reach out to respondents also during non-working hours to improve the likelihood of reaching those who would only be available outside regular working hours. During previous assessments, we collected the respondent's preferred time to contact, which will also be provided for reference if available.

Maximum number of calls

- The maximum number of calls made to the respondent will be 20, spread over a maximum of 6 days. After the 20th attempt, the surveys will drop from the cases. For calls that start on the morning of Day 1, the attempts shall be made as follows:

Day	Attempts	Spread	Contact to use	Who calls?
Day 1	5	Morning, lunchtime, and evening	Primary number	Enumerator
Day 2	0	Skip	Skip	Skip
Day 3	5	Morning, lunchtime, and evening blocks	Run through the numbers including the primary number	Enumerator
Day 4	5	Morning, lunchtime, and evening	Run through the numbers including the primary number	Enumerator
Day 5	0	Skip	Skip	Skip
Day 6	5	Ensure that at least one attempt is on Saturday	Run through the numbers including the primary number	Supervisor

Minimum spacing between calls

- We shall make a maximum of five attempts within a day. The timing of the three calls should be 1) 2 in the morning, 2) lunchtime and 3) 2 in the evening. Here is a breakdown of the hours:
  - Morning: 08:00-12:00
  - Lunchtime: 12:00-15:00
  - Evening: 15:00-20:00
- If the first two attempts are made in the morning, then we should reschedule for 4 hours so that the next attempt is made during lunchtime. The final attempts for that day shall be made approximately 3 hours after the second attempt (ideally one in the afternoon, then once again in the evening).

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- If you make an unsuccessful attempt to reach a respondent just before a skip day, please select a calendar and reschedule the call to the next calling day and choose the morning i.e., 8 am.

Variation in call times for subsequent calls

- Call at lunchtime if the initial call was made in the morning (follow up attempts within the same day should be made at intervals of 4 and 3 hours respectively as outlined above). Call the next morning if the initial call was made in the evening etc.

Dialing protocol

- Let it ring for at least one minute (60 seconds) before hanging up.

When to switch dialed numbers

- All numbers shall be attempted for the maximum number of calls as outlined in this calling protocol even if the number is invalid, busy, or switched off. For complete surveys, they will be dropped from the cases as soon as the survey is successful.
- For refusals, they shall be reassigned to the Supervisor after the first attempt and will only be dropped once the Supervisor has called and spoken to the respondent.

Respondent identification

- Unmatched respondent numbers will be suspended until we confirm that they are our correct respondents. The enumerator should not proceed with the rest of the survey, until they ascertain that this is the correct respondent. If either the name or the age does not match with or even come close to the details we collected in the sample frame, please raise it to the supervisor and suspend the survey.

Reassigning cases

- An enumerator will be able to reassign cases in the following instances:
- Refusals will be reassigned to the Supervisor for her to make another attempt.
- Appointments will be reassigned to other enumerators based on the preferred language of the respondent as per the protocol.

Respondent call backs

- If you receive the automated text message from a number that you tried to reach within the day, or if a respondent calls you back, call immediately and conduct the interview or schedule an appointment instead of conducting an interview at that time if conducting the interview is logistically impossible.

Case escalation

- The Supervisor will respond to the interviewer's escalating issues and escalate any difficult case to the research associate if necessary.

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